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COMMENT



## The two-lane road to hell is paved with good intentions: why an all-or-none approach to generative AI, integrity, and assessment is insupportable

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### ABSTRACT

A 'two-lane' (All-or-None) approach to the use of generative artificial intelligence (genAI) is the idea that there should be two categories of assessments in higher education: Lane 1/None: where the use of genAI is prohibited, and Lane 2/All: where *any* use of genAI is permitted. This idea has been thoughtfully detailed and continues to be debated. Although this idea is generally well-intentioned, in this comment piece I argue that, if implemented, it will promote an impoverished approach to education and educational assessment. One argument often invoked in favour of an All-or-None approach is that genAI use may sometimes be undetectable. Contract cheating (e.g., students outsourcing assessments to ghostwriters) is sometimes undetectable, yet an argument that there should be an All-or-None approach permitting contract cheating in some assessments is clearly absurd. An All-or-None approach to genAI and assessment is also absurd. A middle lane, where genAI use in assessments is allowed with some limitations, is essential.

### ARTICLE HISTORY

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The emergence of ChatGPT and other generative artificial intelligence (genAI) programs has focused the minds of many academics and administrators on the problems of student cheating, assessment design, and assessment validity. One of the proposed ways forward for higher education assessment in the age of genAI has been called the 'two-lane approach' (Liu & Bridgeman, 2023). As it was originally envisioned, in the two-lane approach, the use of genAI tools in assessment is forbidden in 'Lane 1', and these assessments must have a high degree of assessment security, for example, supervised exams. In 'Lane 2', students can use genAI in *any* way to complete *any* component of their *unsecured* assessments.

Scholars who are familiar with the two-lane approach continue to debate it and propose alternatives (e.g., Perkins et al., 2024a; Steel, 2024) and, as an active researcher on academic integrity, this piece is my first contribution to this debate. More anecdotally, however, I have encountered endorsement of the same general position as the two-lane approach from some academics across a range of institutions and disciplines when discussing the issues associated with genAI and academic integrity. The typical argument that I have

encountered is that an All (Lane 2) or None (Lane 1) approach is necessary because students' use of genAI in assessments is 'undetectable'. And, I will use terms All vs None to describe the dichotomy between assessment where genAI is unlimited vs banned.

To be fair to Liu and Bridgeman (2023), their proposal of a two-lane approach is well-argued. They contend that this approach should encourage educators to: deeply rethink assessment design to allow for the existence of genAI; consider assessing the processes of learning and completing the assessment, not just the outcome of assessment; and to rethink grading to focus on higher-order thinking skills. These are all aims that I agree with. The arguments for this position appear to have several bases, some of which are well-intentioned. The good intentions appear, in relation to Lane 2/All, to be: (1) that students will have unlimited ability to use genAI in the workplace, so allowing this in their education sets them up for future work, and (2) that detection is, at best, resource intensive and therefore places a burden on educators that could be removed by allowing students free reign to use genAI. But, as the saying goes, the road to Hell is paved with good intentions, and I believe that going down this two-lane road may have devilish consequences.

The two-lane approach has been both praised and criticised for its simplicity. As Steel (2024) said 'The two-lane approach is brilliant at focusing the mind. But, like many binary options, it over-simplifies'. More importantly, the more naïve, less nuanced, argument for an All-or-None approach often hinges on students' use of genAI in assessment being impossible to prevent and difficult to detect. Neither the argument that it is difficult to prevent or detect ultimately justify the costs of placing no limitations on students' use of genAI in some educational assessments.

To be clear, in arguing against an All-or-None approach to genAI and assessment, I am not arguing for purely using 'None' as an alternative (i.e., no genAI in a setting of high-security, identity-verified, in-person assessment). A preferable approach to genAI, academic integrity, and assessment mixes secure (None) assessment applied programmatically (Lodge, 2024), assessment where genAI is permitted to be used in any way (All), *and* assessments where some limitations are imposed on the use of genAI (which could be called Lane 3 or Some). This approach is consistent with ideas such as the Artificial Intelligence Assessment Scale (Perkins et al., 2024a).

Importantly, where rules are set that limit, but do not prohibit, the use of genAI, detection of breaches of those rules must be attempted. Failure of a student to comply with the limitations placed on using genAI in their assessment may be a case of them making a mistake, which can be corrected educationally, or engaging in cheating, which may result in some other consequence within their higher education institution. Enforcing limitations can readily draw on multi-layer approaches to reduce academic misconduct such as the Swiss Cheese model (Rundle et al., 2020) and the Educational Integrity Enforcement Pyramid (Ellis & Murdoch 2024).

By focusing on how educators can best *assess*, an All-or-None approach potentially undermines how educators can best *teach*. This is particularly true for assessments that permit unlimited genAI use. In the following sections, I will explain why an All-or-None approach undermines good educational and assessment practices, and why the arguments that genAI use cannot be prevented or detected are insufficient to justify an All-or-None approach.

## Education vs assessment

Incremental, scaffolded, and part-learning have always been a feature of good educational practice and *authentic learning* – which should not be confused with *authentic assessment*. Take the example of learning to drive a car. Novice drivers in many jurisdictions must first learn road rules before they can begin practical driving instruction. Practical lessons may begin in dual-control vehicles where the instructor operates breaking and acceleration while the novice driver learns to steer. As novice drivers progress, they may be subject to more constraints than fully licensed drivers, such as lower speed limits, and a requirement of ongoing supervision. Unlimited genAI use (Lane 2/All), if applied to students in their early stages of higher education, is akin to handing a teenager the keys to a high-powered sports car and saying ‘have fun’. This approach ignores the fundamental need to educate students to build skills incrementally within sensible limits.

Students can reasonably expect to have their learning assessed on what they have been *taught*. Thus, if educators expect students to use genAI in completing their assessments, they must show them how. And, as I have argued already, instruction must be incremental to be effective. In this case, the optimal educational strategy of gradually teaching genAI skills to students should be accompanied by their carefully limited use within assessments.

A further educational problem with the All-or-None approach is that allowing unlimited genAI use may decrease students’ academic engagement and learning. Imagine asking a student to write an essay on ‘the theme of death in Hamlet’. Before genAI, the student would have had to read and then write – skills requiring thought, reflection, and time. This task represents a process or authentic learning as people learn deeply by learning gradually. True enough, if the student could find a pre-existing essay on the topic, they could copy it and if they could not, they could outsource it to a ghostwriter. Neither of these options would result in engagement or learning, and both would be considered unacceptable academic misconduct at any higher-education institution. If Lane 1/None is used, the student may write the essay in a time-limited supervised exam without access to genAI. The problem with this is that it does not allow the student to gradually develop their ideas as they learn to read and think about the topic over time. But, if unlimited genAI is permitted, the student can put the question into a genAI program, paste the answer into a document, and submit that document with their name on it (*perhaps* acknowledging the genAI program), and this would, presumably, be acceptable. Like plagiarising or contracting out the assignment, or only writing it under exam conditions, this would result in little academic engagement or learning. Admittedly, sophisticated advocates of the two-lane approach like Liu and Bridgeman (2023) advise against assessments such as this Hamlet example that could be easily completed with genAI. However, how could a higher education institution adopting an All-or-None approach guarantee that its teachers would set assessments that could not easily be completed with emerging genAI technologies? Moreover, when no limits exist, students with access to better AI tools would have an unfair advantage over those who do not.

Unfair advantage, in relation to differing access to technology, is a clear argument against an All-or-None approach. Differing financial resources among students means that some have better access to premium genAI technologies. Unlimited genAI use in

assessments potentially exacerbates the impact of such financial (dis)advantages, where some students complete their assessments more easily, or to a better standard, because of their access to better technology. An obvious solution to this is to have, and enforce, the same limitations on the use of genAI in assessment for all students rather than to have no limitations at all.

Continuing with the Hamlet essay example, a sensible limitation or rule (i.e., Lane 3/Some) for first-year students might be that genAI can be used to research or brainstorm ideas, but not to write the assignment. Advocates of the All-or-None approach would ask: how can we prevent students from breaking this rule and/or how can we detect when they do? I deal with the prevention and detection arguments in the next sections. But, beyond these, my answer, in part, is that we can expect some, many, and, at times, most students to follow assessment rules when they are told what those rules are. A long list of academic integrity studies shows that most students want to learn and will try to complete their assessments within the rules (Curtis & Clare, 2024). Most societies, organisations, and social systems function by people voluntarily following the rules.<sup>1</sup>

However, it is also true that for some students knowing the rules is not enough (Stephens, 2018). For these students, the range of assessment security measures to dissuade rule-breaking and enforce rule compliance are many, varied, and evolving (Birks & Clare, 2023).

### If we can't prevent students using genAI shouldn't we just allow them to use it?

Arguments as to how to prevent contract cheating (i.e., students outsourcing their educational assessments to third parties; Eaton et al., 2022) are well-worn in the literature. Many of the same strategies that educators have tried, and failed, to use to *prevent* contract cheating also fail to prevent genAI use (e.g., authentic assessment and reflective writing; Ellis et al., 2020; Miller, 2024). Nonetheless, worthwhile assessments such as authentic assessment and reflective writing, which may be outsourced, are retained in higher education despite these aspects of assessment design failing to prevent contract cheating. And yet, some argue that assessments should be abandoned where genAI use cannot be prevented, even as the same arguments would likely not be made about assessments that fail to prevent contract cheating.

Dawson et al. (2024), for example, suggest that 'assessments that depend on students not using artificial intelligence but are incapable of *preventing* students from doing so, are not particularly useful for high-stakes assessment of learning' (p. 1012, emphasis added). Educators cannot design assessments, some of which are extremely high stakes (e.g., a dissertation), that *prevent* students from outsourcing these to ghostwriters. Nevertheless, higher education institutions continue using these assessments although they cannot prevent contract cheating. Similarly, earlier I used the analogy of someone learning to drive a car to emphasise that learning is often incremental and assessed under conditions with some limitations rather than without limitation. Continuing the same analogy, a novice driver cannot easily be *prevented* from breaking speed limits but attempts at detecting breaches of the rules are made and these are enforced after breaches occur. Nonetheless, the inability to prevent a breach of the rules does not stop the rule from being set because the rule has value and serves a purpose.

## Contract cheating and the detection argument

A common argument for the All-or-None approach is that genAI use cannot be reliably detected. Drawing an analogy between this position on genAI and holding the same position regarding contract cheating, or plagiarism before the advent of text-matching software, exposes the argument as highly problematic if not downright fallacious.

Some years ago, contract cheating was thought to be impossible to prove, and this is an attitude some people in higher education still hold (Ellis et al., 2022). However, substantial research and innovation in both assessment security and policy has established that a great deal of contract cheating can be successfully detected with dedicated time and resources applied to the problem (Ellis & Murdoch, 2024). Imagine, if you will, in the face of contract cheating being considered impossible to prove, a university institutes a policy that there will be two ‘modes’ of assessment. In Mode 1, students must produce identification, be assessed in person, and must be supervised completing their assessment. In Mode 2, students can recruit any person they want to complete the assessment, put their name on it, and submit this as if it were their work. I am yet to find anyone working in higher education who believes that this idea is acceptable. Why? Because there are many reasons why cheating, and not doing the work of learning, are problems.

When students pass by cheating it devalues degrees as a signal that their holders possess knowledge and competencies, and this undermines the reputation of institutions and graduates (Dawson et al., 2024). Cheating also creates an unfair advantage for one student over another in higher education. It has been argued, in the context of the widespread use of standards-based assessment in Western higher education, that because one student cheating does not affect the grades of another student, cheating does not convey an unfair advantage (Bouville, 2009; Dawson et al., 2024). However, this argument fails to consider the fact that grades, whether achieved in a standards-based system or ‘on a curve’, serve as a currency that is used to determine access to limited resources such as post-graduate places, scholarships, and jobs. A student who cheats to receive higher grades than a student who followed the rules may certainly receive an unfair advantage in competition for the benefits associated with those grades and curtail the opportunities earned by the other student.

As stated earlier, following the rules does not just require advocating that rules should be followed, as this will work for some, but not for all, students. Enforcement is required when rules are broken (Ellis & Murdoch, 2024; Rundle et al., 2020). Detection is necessary to enforce rules related to subverting any genAI limitations placed on assessments, and this is where many in higher education throw up their hands in despair. Like contract cheating, genAI may be considered as ‘undetectable’ by academics who now have a quarter of a century of experience with text-matching software as the principal means of detecting academic misconduct. Text-matching software reliably links text within students’ work to published sources, saving academics countless hours of searching books, websites, and articles to match text in their students’ work that may appear unoriginal. As Weber-Wulff (2009) pointed out, text-matching software has served educators as a crutch in detecting misconduct. Some educators have likely become too accustomed to the ease with which text-matching software can potentially detect plagiarism, and the absence of an equally reliable and efficient method of detecting illicit use of contract cheating or genAI makes the problem seem all too hard. However, I am old enough to

remember the challenge of detecting plagiarism before text-matching software. Despite detection of plagiarism being both difficult and imperfect in that era, no one proposed allowing unlimited plagiarism as a solution to the detection problem. Cheating detection has always been a co-evolutionary arms race (Curtis & Vardanega, 2016) and expecting, or requiring, flawless detection is letting the perfect be the enemy of the good.

GenAI ‘detector’ programs have been created with the aim of providing something functionally equivalent to text-matching software to indicate students’ illicit use of genAI in assessments. These programs are imperfect – their worst feature being false positives – but they may provide probative evidence to detect genAI use. To be clear, a false positive is where a program flags text as being written by genAI when it was not. The most significant concern with such false positives is that if the ‘detector’ is used as the only source of evidence, a student may be put through the unjust and stressful situation of facing a false accusation of academic misconduct. However, many educational institutions have quickly adapted to this problem by, for example, requiring multiple sources of evidence to initiate an investigation of a student for illicit genAI uses rather than relying solely on detectors.

Furthermore, evidence continues to amass that genAI detectors are better at detecting 100% AI generated content than are humans (e.g., Perkins et al., 2024b; Weber-Wulff et al., 2023). Such findings suggest that these programs offer some help and, at least, partial evidence that a student *may* have used genAI in situations where they have been instructed not to. As is done with contract cheating, incomplete probative evidence is combined with other incomplete probative evidence to establish, on the balance of probabilities, whether a student has cheated. For example, a submitted assessment producing a high genAI ‘score’ in a detector program may be more compellingly shown *not* to be a false positive if, for example, document properties and metadata indicate that: the text was pasted rather than typed (e.g., Johnson & Davies, 2020); the document editing time was unusually short; the student cannot show the sources used to write the assessment; and, when asked to discuss the assignment the student fails to demonstrate an understanding of the work.

Lanes 1 and 2 distinguish between secure and unsecure assessment, but assessment security is a continuum not a dichotomy. A promising approach to assessment security in the age of genAI is to use write-in assignment platforms, including commercial products such as Cadmus, which can allow markers to see the process of students’ writing, view their drafts, and quickly distinguish pasted text from typed text. Like genAI detectors, such enhanced assessment security neither prevents nor perfectly detects, the illicit use of genAI. However, this is where the Swiss Cheese Model applied to academic integrity is a reasonable concept to invoke; in short, prevention can be created by layering a series of obstacles, each one having a few ‘holes’, to create a *nearly* impenetrable barrier against undetected misconduct. As with contract cheating, this approach to detection will not catch all students who break rules limiting genAI use in assessments, but catching *some* rule breakers dissuades many others who might contemplate breaking the rules.

## Conclusion

The two-lane approach to genAI, assessment, and academic integrity was conceived with the good intention of helping students prepare for a genAI workplace and lifting the

difficult burden from educators of enforcing limited genAI use. However, an All-or-None approach ignores the purpose of many higher education assessments in gradually and sequentially building knowledge and skills by not allowing for middle-ground assessments where genAI use is permitted within limits and assessment security is partial. It therefore risks throwing away assessments that allow students to develop their understanding gradually through research, thinking, reflection, and revision of their ideas. Consequently, this approach potentially undermines the efficacy and validity of assessment as a motivator of learning. It also undervalues the need for students to build skills, a discipline-specific knowledge base, and evaluative judgement capabilities that are needed to use genAI effectively (Bearman et al., 2024). Worse still, an All-or-None approach seems to be rooted in a defeatist attitude to the prospect of asking students to act with integrity in their assessments and to detect when they have not. As Corbin et al. (2025) recently stated, ‘policies for AI ought to move beyond simple prohibitions or permissions ... What counts as appropriate AI use varies not just between disciplines but between different types of learning tasks and outcomes’. I agree.

## Note

1. Thanks to Joe Clare for this insight.

## Disclosure statement

No potential conflict of interest was reported by the author(s).

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